United States Patent	12383875
Kind Code	B2
Date of Patent	August 12, 2025
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Static mixer

Abstract

A static mixer including a first inlet channel, a second inlet channel, and a first dividing wall between the first inlet channel and the second inlet channel. The static mixer further includes a first outlet channel aligned with the first inlet channel along a first axis and a second outlet channel aligned with the second inlet channel along a second axis. The static mixer further includes a fin extending from the dividing wall.

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Appl. No.:	**17/770698**
Filed (or PCT Filed):	**October 21, 2020**
PCT No.:	**PCT/US2020/056706**
PCT Pub. No.:	**WO2021/081122**
PCT Pub. Date:	April 29, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220410092 A1	Dec. 29, 2022

Related U.S. Application Data

us-provisional-application US 62924609 20191022
us-provisional-application US 62924170 20191021

Publication Classification

Int. Cl.: **B01F25/432** (20220101)

U.S. Cl.:

CPC **B01F25/4321** (20220101);

Field of Classification Search

CPC: B01F (25/431); B01F (25/4313); B01F (25/432); B01F (25/4321)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a 371 U.S. National Phase Entry of pending International Application No. PCT/US2020/056706, filed Oct. 21.2020, which claims priority to U.S. Provisional Patent Application No. 62/924,609 filed on Oct. 22, 2019 and U.S. Provisional Patent Application No. 62/924,170 filed on Oct. 21, 2019, the entire contents of which are incorporated herein by reference.

FIELD OF THE INVENTION
(1) The present disclosure relates to a static mixer.
BACKGROUND OF THE INVENTION

(2) A number of conventional motionless (i.e., static) mixer types exist that implement a similar general principle to mix fluids together. Specifically, fluids are mixed together by dividing and recombining the fluids in an overlapping manner. This action is achieved by forcing the fluid over a series of baffles of alternating geometry. Such division and recombination cause the layers of the fluids being mixed to diffuse past one another, eventually resulting in a generally homogenous mixture of the fluids. However, conventional mixers often result in a streaking phenomenon with streaks of fluid that pass through the mixer essentially unmixed.
(3) Furthermore, to achieve adequate mixing (i.e., a generally homogenous mixture) additional baffles must be placed in the conventional mixer to thoroughly diffuse the material, thus increasing the mixer's overall length. Such an increase in mixer length is unacceptable in many motionless mixer applications, such as handheld mixer-dispensers. In addition, longer mixers generally have a higher retained volume and higher amounts of waste material as a result. A large amount of waste material is particularly undesirable when dealing with expensive materials. In other words, the length of the conventional static mixer is large, resulting in a large amount of wasted material that must pass through the static mixer before any mixed output is usable.
SUMMARY OF THE INVENTION
(4) The disclosure provides, in one aspect, a static mixer including a first inlet channel, a second inlet channel, and a first dividing wall between the first inlet channel and the second inlet channel. The static mixer further includes a first outlet channel aligned with the first inlet channel along a first axis and a second outlet channel aligned with the second inlet channel along a second axis. The static mixer further includes a fin extending from the dividing wall.
(5) The disclosure provides, in another aspect, a static mixer including a first channel, a second channel, a third channel, and a first dividing wall positioned between the first channel and the second channel. The static mixer further includes a second dividing wall positioned between the second channel and the third channel. A first fin extends from the first dividing wall and a second fin extends from the second dividing wall.
(6) The disclosure provides, in another aspect, a static mixer including a first channel, a second channel, a third channel, and a first dividing wall between the first channel and the second channel. The static mixer further includes a second dividing wall between the second channel and the third channel. A first opening is formed in the first dividing wall and in fluid communication with the first channel, and a second opening is formed in the second dividing wall and in fluid communication with the second channel. The first dividing wall includes a first flange at least partially defining the first opening and the second dividing wall includes a second flange at least partially defining the second opening. The first opening and the second opening are not the same size.
(7) The disclosure provides, in another aspect, a static mixer including a first channel at least partially defined by a guide wall, a second channel, and a dividing wall positioned between the first channel and the second channel. The guide wall has a first thickness and a second thickness different than the first thickness.
(8) The disclosure provides, in another aspect, a static mixer including a first channel defining a first dimension, a second channel defining a second dimension, and a third channel. The static mixer further includes a first dividing wall between the first channel and the second channel, and a second dividing wall between the second channel and the third channel. The first dimension is less than half the second dimension.
(9) Other aspects of the disclosure will become apparent by consideration of the detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a side view of a static mixer according to an aspect of the disclosure.
(2) FIG. **2** is an exploded view of the static mixer of FIG. **2** illustrating a mixer assembly.
(3) FIG. **3** is a perspective view of mixer assembly of FIG. **2**.
(4) FIG. **4** is a side view of the mixer assembly of FIG. **3**.
(5) FIG. **5** is an enlarged partial perspective view of a mixer assembly according to an aspect of the disclosure.
(6) FIG. **6** is a front view of the mixer assembly of FIG. **3**.
(7) FIG. **7** is an enlarged partial top view of the mixer assembly of FIG. **3**.
(8) FIG. **8** is a cross-sectional view of the mixer assembly of FIG. **3**, taken along lines **8-8** shown in FIG. **7**.
(9) FIG. **9** is a perspective view of a mixer element according to another aspect of the disclosure.
(10) FIG. **10** is a perspective view of a mixer element according to another aspect of the disclosure.
(11) FIG. **11** is a perspective view of a mixer element according to another aspect of the disclosure.
(12) FIG. **12** is a perspective view of a mixer element according to another aspect of the disclosure.
(13) FIG. **13** is a front view of the mixer element of FIG. **12**.
(14) FIG. **14** is a partial perspective view of a mixer assembly according to another aspect of the disclosure.
(15) FIG. **15** is a partial side view of the mixer assembly of FIG. **14**.
(16) Before any embodiments of the disclosure are explained in detail, it is to be understood that the invention is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The invention is capable of other embodiments and of being practiced or of being carried out in various ways.

DETAILED DESCRIPTION

(17) With reference to FIGS. **1** and **2**, a static mixer **10** according to one embodiment of the invention is illustrated. The static mixer **10** includes a housing **14** and a mixer assembly **18** received within the housing **14**. Specifically, the housing **14** includes an inlet end **22** formed with an inlet socket **26** and an outlet end **30** formed with a nozzle **34**. The inlet end **22** and the outlet end **30** define a material flow path that extends therebetween. In other words, the inlet end **22** is upstream in the material flow path from the outlet end **30**. In the illustrated embodiment, the inlet socket **26** is formed as a bell-type inlet, but in alternative embodiments the inlet socket **26** may be formed as a bayonet-type inlet, for example. Other inlet configurations known to those of ordinary skill in the art may also be used.
(18) With continued reference to FIG. **2**, the static mixer **10** includes an overall length **38**, which is smaller than the overall length of conventional static mixers. As explained in greater detail below, the static mixer **10** is able to create a more homogenous mixture (i.e., improved results) with a shorter overall length (i.e., less wasted material) compared to conventional mixers. A static mixer is disclosed in U.S. patent application Ser. No. 15/526,556, the entire contents of which are incorporated herein by reference.
(19) With reference to FIG. **2**, the mixer assembly **18** is received within a chamber **42** (i.e., channel) defined by the housing **14**. In the illustrated embodiment, the chamber **42** is square-shaped with four chamber walls **46**. In alternative embodiments, the chamber **42** may be circular-shaped to correspond to a circular-shaped mixer element. The mixer assembly **18** includes seven mixer elements **50**A, **50**B, **50**C, **50**D, **50**E, **50**F, **50**G. As explained in greater detail below, two or more separate fluids (e.g., gasses, liquids, and/or fluidized solids) enter the inlet end **12** of the housing **14**, pass through the mixer assembly **18** and exit through the outlet end **30** as a substantially homogenous mixture.
(20) The mixer assembly **18** can be formed by the combination of mixer elements with various

geometries in various orientations. The mixer assembly **18** is illustrated with seven mixer elements **50**A-**50**G and are referenced sequentially in a downstream direction. The second mixer element **50**B is positioned downstream in the material flow path from the first mixer element **50**A. The third mixer element **50**C is positioned downstream in the material flow path from the second mixer element **50**B. The fourth mixer element **50**D is positioned downstream in the material flow path from the third mixer element **50**C. In the illustrated embodiment, the seven mixer elements **50**A-**50**D are formed as a single integral unit (i.e., formed with an injection molding process or 3D printing process). In some embodiments, the mixer assembly is formed by a plurality of mixer elements with the same, or similar, geometry. With continued reference to FIGS. **3-4**, the second, third, and fourth mixer elements **50**B, **50**C, **50**D are the same structure illustrated for the first mixer element **50**A. However, the third mixer element **50**C is positioned in a different orientation as the second mixer element **50**B and the fourth mixer element **50**D is positioned in a different orientation as the third mixer element **50**C. In other words, the mixer assembly **18** defines a longitudinal axis **54** and the mixer elements **50**A-**50**G are positioned in different orientations rotationally about the longitudinal axis **54**. For example, the second mixer element **50**B is oriented with a 90 degree rotation along the longitudinal axis **54** with respect to the first mixer element **50**A, and the third mixer element **50**C is oriented with a 90 degree rotation along the longitudinal axis **54** with respect to the second mixer element **50**B. In some embodiments, mixer elements with different geometries are combined to formed a mixer assembly. Details and aspects of various mixer elements (e.g., FIGS. **5-8**; FIG. **9**; FIG. **10**; FIG. **11**; FIGS. **12-13**; and FIGS. **14-15**) are discussed below.

(21) With reference to FIGS. **5-8**, a mixer element **58** includes six inlet channels **62**A, **62**B, **62**C, **62**D, **62**E, **62**F and six outlet channels **68**A, **68**B, **68**C, **68**D, **68**E, **68**F. The inlet channels **62**A-**62**F are upstream in a material flow path of the outlet channels **68**A-**68**F. Each of the outlet channels **68**A-**68**F is aligned with a corresponding inlet channel **62**A-**62**F along a channel axis **72**A, **72**B, **72**C, **72**D, **72**E, **72**F. For example, the first outlet channel **68**A is aligned with the first inlet channel **62**A along the first axis **72**A, and second outlet channel **68**B is aligned with the second inlet channel **62**B along the second axis **72**B. In addition, third outlet channel **68**C is aligned with the third inlet channel **62**C along the third axis **72**C, and so forth. The first axis **72**A is approximately parallel with the second axis **72**B. In the mixer element **58** of FIGS. **5-8**, the axes **72**A-**72**F are parallel with each other. The mixer element **58** is substantially the same as mixer elements **50**A-**50**G illustrated in the mixer assembly **18**.

(22) With continued reference to FIG. **5**, the mixer element **58** further includes a first set of openings **76**A, **76**B, **76**C, **76**D, **76**E and a second set of openings **80**A, **80**B, **80**C, **80**D, **80**E. With reference to FIG. **5** and its frame of reference, the first set of openings **76**A-**76**E are upper openings and the second set of openings **80**A-**80**E are lower openings. In particular, the five openings **76**A-**76**E are positioned between the inlet channels **62**B, **62**D, **62**F and the outlet channels **68**A, **68**C, **68**E. Similarly, the five openings **80**A-**80**E are positioned between the inlet channels **62**A, **62**C, **62**E and the outlet channels **68**B, **68**D, **68**F. Specifically, the first opening **76**A is between the second inlet channel **62**B and the first outlet channel **68**A, and the second opening **76**B is between the second inlet channel **62**B and the third outlet channel **68**C. In other words, the openings **76**A-**76**E and **80**A-**80**E place an inlet channel **62**A-**62**F in fluid communication with an adjacent one of the outlet channels **68**A-**68**F (i.e., an outlet channel next to, but not aligned with the inlet channel).

(23) With reference to FIGS. **5** and **7**, the upper openings **76**A-**76**F are all aligned along an upper opening axis **84**, and the lower opening **80**A-**80**E are all aligned along a lower opening axis **88**. In other words, the upper opening axis **84** passes through a centroid of each of the upper openings **76**A-**76**F. Likewise, the lower opening axis **88** passes through a centroid of each of the lower openings **80**A-**80**E. In the embodiment illustrated, the first set of openings **76**A-**76**F are all the same size and the second set of openings **80**A-**80**F are all the same size. In addition, in the illustrated embodiment, the first set of openings **76**A-**76**F are the same size as the second set of openings **80**A-**80**F. In other embodiments, the openings **76**A-**76**F and **80**A-**80**F are different sizes

or different shapes.
(24) With continued reference to FIGS. **5-8**, the mixer element **58** can alternatively be described as including wall segments. The mixer element **58** includes a plurality of guide walls **92**A-**92**F and a plurality of dividing walls **96**A-**96**E extending between the guide walls **92**A-**92**F. Specifically, a first dividing wall **96**A extends between the first guide wall **92**A and the second guide wall **92**B. The first inlet channel **62**A is partially defined by the first guide wall **92**A and the first dividing wall **96**A. The first outlet channel **68**A is also partially defined by the first guide wall **92**A and the first dividing wall **96**A. In other words, the first outlet channel **68**A is positioned on an opposite side of the first guide wall **92**A as the first inlet channel **62**A (i.e., the first guide wall **92**A separates the first inlet channel **62**A and the first outlet channel **68**A). When the mixer element **58** is positioned with the housing **14**, the first guide wall **92**A completely separates the first inlet channel **62**A and the first outlet channel **68**A such that the first inlet channel **62**A is not in fluid communication with the first outlet channel **68**A.
(25) With continued reference to FIG. **5**, the first opening **76**A is at least partially defined by the first dividing wall **96**A. The first opening **76**A places the second inlet channel **62**B in fluid communication with the first outlet channel **68**A. The second opening **80**A is also at least partially defined by the first dividing wall **96**A. The second opening **80**A places the first inlet channel **62**A in fluid communication with the second outlet channel **68**B. In the illustrated embodiment, an outer periphery of the first dividing wall **92**A at least partially defines the first opening **76**A and at least partially defines the second opening **80**A. In the illustrated embodiment, the openings **76**A-**76**E and **80**A-**80**E are triangular-shaped. In some embodiments, the openings **76**A-**76**E and **80**A-**80**E are at least partially formed by a flange **100**. In alternative embodiments, the openings **76**A-**76**E and **80**A-**80**E are curved (i.e., at least partially defined by an arc). With reference to FIGS. **5** and **6**, the first dividing wall **96**A and the second dividing wall **96**B are parallel to each other. In the illustrated embodiment, each of the dividing walls **96**A-**96**E are parallel to each other.
(26) With reference to FIGS. **5** and **7**, the first guide wall **92**A is non-planar (i.e., a curved surface) and the second guide wall **92**B is non-planar (i.e., a curved surface). In other words, the first guide wall **92**A does not extend along a straight line (i.e., the first guide wall **92**A is curve-shaped). Likewise, the second guide wall **92**B does not extend along a straight line (i.e., the second guide wall **92**B is curve-shaped). In the illustrated embodiment, the guide walls **92**A-**92**F have a similar shape. In some embodiments, the guide walls **92**A-**92**F are S-shaped. In other embodiments, the guide walls **92**A-**92**F are sigmoid shaped.
(27) With continued reference to FIGS. **5** and **7**, the first guide wall **92**A defines a variable thickness. Specifically, the first guide wall **92**A has a first thickness **104** at an upstream portion **108** of the guide wall **92**A and a second thickness **112** at a midstream portion **116** of the guide wall **92**A. The first thickness **104** is different than the second thickness **112**. In the illustrated embodiment, the first thickness **104** is larger than the second thickness **112**. In other embodiments, the first thickness **104** is smaller than the second thickness **112**. The guide wall **92**A also defines a third thickness **120** at a downstream portion **124** of the guide wall **92**A. The midstream portion **116** of the guide wall **92**A is positioned between the upstream portion **108** and the downstream portion **124**. In the illustrated embodiment, the upstream portion **108** extends approximately parallel to the downstream portion **124** with the midstream portion **116** extending therebetween in a non-linear manner. In some embodiments, the first thickness **104** is approximately equal to the third thickness **120**. Specifically, the thickness of the guide wall **92**A as used herein is the shortest distance through the guide wall **92**A at any point along the guide wall **92**A. In other words, the guide wall **92**A includes a first surface **128** (i.e., an upstream surface) and a second surface **132** (i.e., a downstream surface). The thickness of the guide wall **92**A is defined as the shortest distance from a given point on the first surface **128** to the second surface **132**. Increasing guide wall thickness as the guide walls approach housing corners reduces the streaking in the resulting mixture of materials.
(28) With reference to FIGS. **5** and **6**, the first inlet channel **62**A is less than half the width of the

second inlet channel **62**B. Specifically, the first inlet channel **62**A defines a first dimension **136** (i.e., width) and the second inlet channel **62**B defines a second dimension **140**. The first dimension **136** is less than half the second dimension **140**. As discussed above, the housing **14** surrounds the mixer element **58** and the first dimension **136** is the distance between the first dividing wall **96**A and a side wall **46** of the chamber **42** in the housing **14**. In other words, the first inlet channel **62**A is an end channel positioned at an outer periphery of the mixer element **58** and the second inlet channel **62**B is an internal channel positioned with between two adjacent channels (i.e., the first inlet channel **62**A and the third inlet channel **62**C). In the illustrated embodiment, the second inlet channel **62**B, the third inlet channel **62**C, the fourth inlet channel **62**D, the fifth inlet channel **62**E are approximately the same width (i.e., the second dimension **140**); while the first inlet channel **62**A and the sixth inlet channel **62**F are approximately the same width (i.e., the first dimension **136**). The first guide wall **92**A partially defines the first inlet channel **62**A and the first guide wall **92**A extends between the first dividing wall **96**A and the housing **14**. The second dimension **140** is the distance between the first dividing wall **96**A and the second dividing wall **96**B. The second guide wall **92**B partially defines the second inlet channel **62**A and the second guide wall **92**B extends between the first dividing wall **96**A and the second dividing wall **96**B. In some embodiments, the first dimension **136** is approximately 40 percent of the second dimension **140**. In other embodiments, the first dimension **136** is less than approximately 40 percent of the second dimension **140**. In some embodiments, the first dimension **136** may be within a range of approximately 35 percent to approximately 45 percent of the second dimension **140**. Proportioning the first dimension **136** accordingly, reduces the streaking in the resulting mixture of materials.

(29) With reference to FIG. **5**, the static mixer **10** includes a plurality of fins **144**A-**144**E, **148**A-**148**E extending from the dividing walls **96**A-**96**E. The fins **144**A-**144**E (i.e., the upstream fins) extend from the upstream end of the dividing walls **96**A-**96**E and the fins **148**A-**148**E (i.e., the downstream fins) extend from the downstream end of the dividing walls **96**A-**96**E. In the illustrated embodiment, the fins **144**A, **144**C, **144**E, **148**A, **148**C, **148**E extend co-planar with the dividing walls **96**A, **96**C, **96**E (e.g., the fin **144**A and the fin **148**A are coplanar with the dividing wall **96**A). In the illustrated embodiment, the fins **144**B, **144**D, **148**B, and **148**D are shaped so as to extend towards and partially block the flow of material to one of the inlet channels **62**A-**62**F or from one of the outlet channels **68**A-**68**F (e.g., the fin **144**B extends towards and partially blocks the flow of material to the inlet channel **62**A; and the fin **148**B extends towards and partially blocks the flow of material from the outlet channel **68**B). In other words, the fins **144**B, **144**D, **148**B, and **148**D extend at angles with respect to the dividing walls **96**B, **96**D and at least partially overlap with one of the inlet channels **62**-**62**F or one of the outlet channels **68**A-**68**F as viewed from an upstream or downstream end view (e.g., FIG. **6**).

(30) With continued reference to FIGS. **5** and **6**, each of the fins **144**A-**144**E and **148**A-**148**E define a first edge surface **152** (i.e., a first edge), a second edge surface **158** (i.e., a second edge), and a third edge surface **162** (i.e., a third edge). The second edge surface **158** and the third edge surface **162** of each fin **144**A-**144**E and **148**A-**148**E are coupled one of the dividing walls **96**A-**96**E. In the illustrated embodiment, the second edge surface **158** and the third edge surface **162** of the fin **144**B, for example, are directly connected to the dividing wall **96**B. The first edge surface **152** extends between the second edge surface **158** and the third edge surface **162**. In the illustrated embodiment, the first edge surface **152**, the second edge surface **158**, and the third edge surface **162** of the fins **144**A-**144**E and **148**A-**148**E are linear and extend in a linear direction. In other words, the first edge surface **152** defines a linear profile and the second and third edge surfaces **158** and **162** define a linear loft. Each fin **144**A-**144**E and **148**A-**148**E includes a first side surface **166** and a second side surface **170** positioned opposite the first side surface **166**. In other words, a first side **174** of the fin **144**A is at least partially defined by the first side surface **166** and a second side **178** of the fin **144**A is at least partially defined by the second side surface **170**. With reference to FIG. **6** and its frame of reference, the second edge surface **158** is an upper edge of the fins **144**A-

144E and the third edge surface **162** is a lower edge of the fins **144**A-**144**E.
(31) With continued reference to FIG. **5**, the first edge surfaces **166** and the second edge surfaces **158** of the fins **144**A-**144**E and **148**A-**148**E are linear. For example, the fin **144**B extends from a terminal edge **182** of the dividing wall **96**B along a linear path that at least partially defines the second edge surface **158**. In other words, the second edge surface **158** extends linearly from the terminal end **182** on the dividing wall **96**B to the first edge surface **166**. The second edge **158** extends from the dividing wall **96**B at an angle. The third edge **162** extends from the dividing wall **96**B along the plane of the dividing wall **96**B. In other words, the third edge surface **162** on the fin **144**B is co-planar with the dividing wall **96**B. In other embodiments, any one of the first edge **152**, the second edge **158**, or the third edge **162** are non-linear.
(32) As illustrated in FIGS. **5**, **7**, and **8**, the downstream fins **148**A-**148**E are coupled to upstream fins **144**A-**144**E of another mixer element **58**B. In the illustrated embodiment, the downstream mixer element **58**B is the same as the upstream mixer element **58** but rotated 90 degrees. In the illustrated embodiment, each of the downstream fins **148**A-**148**E of the mixer element **58** intersects at least two of the upstream fins **144**A-**144**E of the adjacent mixer element **58**B. In some embodiments, each downstream fin **148**A-**148**E of the mixer element **58** intersects each of the upstream fins **144**A-**144**E of the adjacent mixer element **58**B.
(33) In operation of the mixer element **58**, material entering the inlet channels **62**A-**62**F is guided by the guide walls **92**A-**92**F toward the openings **76**A-**76**E and **80**A-**80**E. The material then passes from the inlet channels **62**A-**62**F through the openings **76**A-**76**E and **80**A-**80**E to the outlet channels **68**A-**68**F. Specifically, the material flows from an inlet channel into an adjacent outlet channel through an opening. For example, material entering the inlet channel **62**A is guided by the first guide wall **92**A toward the opening **80**A where the material then enters the second outlet channel **68**B (i.e., an outlet channel adjacent the inlet channel). As such, the first inlet channel **62**A is not in fluid communication with the first outlet channel **68**A. The fins **144**A-**144**E and **148**A-**148**E improves the overall mixing performance of the mixer element **58**. For example, the fins **144**A-**144**E and **148**A-**148**E reduce the amount of streaking that occurs in an output of the mixer element **58**. The fins **144**A-**144**E and **148**A-**148**E also reduce the pressure loss across the mixer element **58**.
(34) With reference to FIG. **9**, a mixer element **182** similar to the mixer element **58** is illustrated with similar reference numerals from the mixer element **58** used to describe the mixer element **182**. Only the differences between the mixer element **182** and the mixer element **58** are described herein. None of the fins **144**A-**144**E and **148**A-**148**E extend co-planar to any one of the dividing walls **96**A-**96**E. In other words, the upstream fins **144**A-**144**E and the downstream fins **148**A-**148**E extend at angles with respect to the dividing walls **96**A-**96**E. The first edge **152**, the second edge **158**, and the third edge **162** are linear. In other words, the fins **144**A-**144**E and **148**A-**148**E have both a linear profile (i.e., the first edge **152** is linear) and a linear loft (i.e., the second and third edge **158**, **162** are linear). In the illustrated embodiment, a portion of the fin **144**A contacts a portion of the fin **144**B. Specifically, a corner **186** (i.e., the intersection of the first edge surface **152** and the second edge surface **158**) on the fin **144**A contacts a corner **190** (i.e., the intersection of the first edge surface **152** and the second edge surface **158**) on the fin **144**B. Likewise, a corner **194** (i.e., the intersection of the first edge surface **152** and the third edge surface **162**) on the fin **144**B contacts a corner **198** (i.e., the intersection of the first edge surface **152** and the third edge surface **162**) on the fin **144**C.
(35) With reference to FIG. **10**, a mixer element **202** similar to the mixer element **58** is illustrated with similar reference numerals from the mixer element **58** used to describe the mixer element **202**. Only the differences between the mixer element **202** and the mixer element **58** are described herein. None of the fins **144**A-**144**E and **148**A-**148**E extend co-planar to any one of the dividing walls **96**A-**96**E. In other words, the upstream fins **144**A-**144**E and the downstream fins **148**A-**148**E extend at angles with respect to the dividing walls **96**A-**96**E. The second edge **158** and the third

edge **162** are linear, and the first edge **152** is non-linear (i.e., curved, arcuate, etc.). In other words, the fins **144**A-**144**E and **148**A-**148**E have a non-linear profile (i.e., the first edge **152** is non-linear) and a linear loft (i.e., the second and third edge **158**, **162** are linear). In the illustrated embodiment, a portion of the fin **144**A contacts and is integrally formed with a portion of the fin **144**B. Specifically, a corner **206** on the fin **144**A contacts and is connected to a corner **210** on the fin **144**B. Likewise, a corner **214** on the fin **144**B contacts and is connected to a corner **218** on the fin **144**C.

(36) With reference to FIG. **11**, a mixer element **222** similar to the mixer element **58** is illustrated with similar reference numerals from the mixer element **58** used to describe the mixer element **222**. Only the differences between the mixer element **222** and the mixer element **58** described herein. None of the fins **144**A-**144**E and **148**A-**148**E extend co-planar to any one of the dividing walls **96**A-**96**E. In other words, the upstream fins **144**A-**144**E and the downstream fins **148**A-**148**E extend at angles with respect to the dividing walls **96**A-**96**E. The first edge **152** is linear and the second edge **158** and the third edge **162** are non-linear (i.e., curved, arcuate, etc., and). In other words, the fins **144**A-**144**E and **148**A-**148**E have a linear profile (i.e., the first edge **152** is linear) and a non-linear loft (i.e., the second and third edge **158**, **162** are non-linear). In the illustrated embodiment, a portion of the fin **144**A contacts and is integrally formed with a portion of the fin **144**B. Specifically, a corner **226** on the fin **144**A contacts a corner **230** on the fin **144**B. Likewise, a corner **234** on the fin **144**B contacts a corner **238** on the fin **144**C. In some embodiments, all of the edges **152**, **158**, **162** of any one of the fins **144**A-**144**E and **148**A-**148**E are non-linear creating fins with both non-linear profiles and non-linear lofts.

(37) With reference to FIGS. **12** and **13**, a mixer element **242** similar to the mixer element **58** is illustrated with similar reference numerals from the mixer element **58** used to describe the mixer element **242**. Only the differences between the mixer element **242** and the mixer element **58** described herein. The upstream fins **144**A-**144**E and the downstream fins **148**A-**148**E extend at angles with respect to the dividing walls **96**A-**96**E. The edges **152**, **158**, **162** of the fins **144**A-**144**E and **148**A-**148**E are linear. Each of the fins **144**A-**144**E and **148**A-**148**E includes multi-planar sides. For example, a first side **246** of the fin **144**A includes a first planar side surface **250** and a second planar side surface **254**. A portion of the second planar surface **254** is positioned between the first planar side surface **250** and the dividing wall **96**A. Likewise, a second side **258** of the fin **144**A includes a first planar side surface **262** and a second planar side surface **266**. In other words, each of the sides **246** and **258** of the fin **144**A includes more than one planar surface (i.e., first and second planar side surfaces **250**, **254** and **262** and **266**).

(38) With reference the various embodiments of FIGS. **5-12**, the shape and geometry of the fins **144**A-**144**E and **148**A-**148**E are optimized to provide improved mixing results. Added curvature to the profile and loft of the fins **144**A-**144**E and **148**A-**148**E can result in reduced streaking and a reduction in the total pressure loss across the mixer element, respectively.

(39) With reference to FIGS. **14** and **15**, a mixer assembly **270** is illustrated with a first mixer element **274**A and a second mixer element **274**B. The mixer element **274**A is similar to the mixer element **58** and is illustrated with similar reference numerals from the mixer element **58** used to describe the mixer element **274**A. Only the differences between the mixer element **274**A and the mixer element **58** described herein. The mixer element **274**A includes a plurality of inlet channels **62**A-**62**F including a first inlet channel **62**A, a second inlet channel **62**B, and a third inlet channel **62**C. The mixer element **274**A also includes corresponding outlet channels **68**A-**68**F aligned with the inlet channels **62**A-**62**F. A first dividing wall **96**A is positioned between the first inlet channel **62**A and the second inlet channel **62**B and a second dividing wall **96**B is positioned between the second inlet channel **62**B and the third inlet channel **62**C.

(40) The mixer element **274**A includes a first set of openings **278**A-**278**E (i.e., upper openings) and a second set of openings **282**A-**282**E (i.e., lower openings). The first opening **278**A is formed in the first dividing wall **96**A. The first opening **278**A is in fluid communication with the first inlet

channel **62**A. Specifically, the first opening **278**A is positioned between and fluidly communicates the first inlet channel **62**A and the second outlet channel **68**B. The second opening **278**B is formed in the second dividing wall **96**B. The second opening **278**B is in fluid communication with the second inlet channel **62**B. Specifically, the second opening **278**B is positioned between and fluidly communicates the second inlet channel **62**B and the third outlet channel **68**C. The first opening **278**A and the second opening **278**B are not the same size. In the illustrated embodiment, the first opening **278**A is larger than the second opening **278**B. In addition, the second opening **278**B is larger than the third opening **278**C. In the illustrated embodiment, the second opening **278**B is the same size as the fourth opening **278**D and the first opening **278**A is the same size as the fifth opening **278**E. The first opening **278**A and the second opening **278**B are aligned along an axis **84**. In the illustrated embodiment, the axis **84** passes through all the upper openings **278**A-**278**E and a second axis **88** passes through all the lower openings **282**A-**282**E.

(41) With continued reference to FIGS. **14** and **15**, the first dividing wall **96**A includes a flange **100** that at least partially defines the first opening **278**A. The second dividing wall **96**B also includes a flange **100** that at least partially defines the second opening **278**B. The first opening **278**A and the second opening **278**B are triangular shaped. Each of the flanges **100** includes two linear portions **101**, **102** that are themselves triangular shaped. In some embodiments, the flange **100** is non-linear. In other embodiments, the first opening **278**A and the second opening **278**B are not the same shape or the flanges **100** that at least partially define the openings **278**A, **278**B are different shapes.

(42) Although the disclosure has been described in detail with reference to certain embodiments above, variations and modifications exist within the scope and spirit of one or more independent aspects of the disclosure as described.

Claims

1. A static mixer comprising: a first inlet channel; a second inlet channel; a dividing wall between the first inlet channel and the second inlet channel; a first outlet channel aligned with the first inlet channel along a first axis; a second outlet channel aligned with the second inlet channel along a second axis; wherein the dividing wall is between the first outlet channel and the second outlet channel; an opening formed in the dividing wall; wherein the opening is in fluid communication with the first inlet channel and the second outlet channel; and a fin extending from the dividing wall; wherein the fin partially blocks a flow of material to the first inlet channel or the fin partially blocks a flow of material from the first outlet channel.
2. The static mixer of claim 1, wherein the fin includes a first edge and a second edge, the second edge is coupled to the dividing wall.
3. The static mixer of claim 2, wherein the fin includes a third edge coupled to the dividing wall.
4. The static mixer of claim 2, wherein the first edge is linear and the second edge is linear.
5. The static mixer of claim 2, wherein the first edge is linear and the second edge is non-linear.
6. The static mixer of claim 2, wherein the first edge is non-linear and the second edge is linear.
7. The static mixer of claim 2, wherein the first edge is non-linear and the second edge is non-linear.
8. The static mixer of claim 1, wherein the fin defines a side with a first planar surface and a second planar surface.
9. The static mixer of claim 1, further including a guide wall at least partially defining the first inlet channel.
10. The static mixer of claim 9, wherein the guide wall is non-planar.
11. The static mixer of claim 10, wherein the guide wall has a first thickness and a second thickness different than the first thickness.
12. The static mixer of claim 11, wherein an upstream portion of the guide wall defines the first thickness and a midstream portion of the guide wall defines the second thickness; and wherein the

first thickness is larger than the second thickness.
13. The static mixer of claim 1, wherein the opening is at least partially defined by a flange.
14. A static mixer including a first inlet channel; a second inlet channel; a third inlet channel; a first dividing wall positioned between the first inlet channel and the second inlet channel; a second dividing wall positioned between the second inlet channel and the third inlet channel; a first fin extending from the first dividing wall; and a second fin extending from the second dividing wall; wherein the first fin partially blocks a flow of material to the first inlet channel; and wherein the second fin is co-planar with the second dividing wall.
15. The static mixer of claim 14, further comprising a first outlet channel aligned with the first inlet channel along a first axis; a second outlet channel aligned with the second inlet channel along a second axis; and a third outlet channel aligned with the third inlet channel along a third axis; and wherein the first dividing wall is between the first outlet channel and the second outlet channel; and wherein the second dividing wall is between the second outlet channel and the third outlet channel.
16. The static mixer of claim 15, further comprising a third fin extending from the first dividing wall; and a fourth fin extending from the second dividing wall.
17. The static mixer of claim 16, wherein the third fin partially blocks a flow of material from the first outlet channel; and wherein the fourth fin is co-planar with the second dividing wall.
18. The static mixer of claim 17, wherein the first fin includes a first edge, a second edge, and a third edge; wherein the second edge and the third edge are coupled to the first dividing wall; and wherein the first edge, the second edge, and the third edge are linear.
19. The static mixer of claim 14, further comprising a first opening formed in the first dividing wall and in fluid communication with the first inlet channel; and a second opening formed in the second dividing wall and in fluid communication with the second inlet channel; and wherein the first opening and the second opening are not the same size.
20. The static mixer of claim 14, further comprising a housing; and wherein the first inlet channel defines a first dimension that is the distance between the first dividing wall and the housing; and the second inlet channel defines a second dimension that is the distance between the first dividing wall and the second dividing wall; and wherein the first dimension is less than half the second dimension.